

3. On the Invariant Theory of Forms in n Variables¹

Jahresbericht der DMV 19 (1910), pp. 101–104

In the projective invariant theory of forms in n variables the main problem has been solved: the finiteness of the system of forms has been proved. A series of further questions, however, have not been treated, namely those concerned with methods for investigating the connection among forms. I would like briefly to communicate the results that I have obtained with respect to such questions, but for the sake of clarity I shall first sketch the questions in the ternary domain, where they are known.

I have first in mind the two fundamental theorems of the symbolic method: the theorem that all invariant formations can be represented symbolically, and the second theorem that all relations existing among invariants are obtained by successive application of a small number of symbolic identities; thus the symbolic method gives all relations without leaving the domain of invariants.² Built upon this are the processes for generating the forms, the symbolic folding process and the nonsymbolic differentiation processes, the theory of series expansions, and the like. In this connection I should also mention a theorem proved in my dissertation,³ namely that all foldings can be generated by successive application of the two possible “cogredient” foldings; by this, for a symbolic product $s_x t_x u_\sigma u_\tau$, I mean the two foldings (stu) and $(\sigma\tau x)$. On this theorem one can build the theory of reducers and of the reduction of systems of forms, in analogy with the binary domain, as I showed in my dissertation.

When we now pass to the n -ary domain, even the first theorem of the symbolic method holds only in a conditional measure. It is well known that already for quaternary forms there occur, besides the point and plane coordinates x and u , line coordinates p_{ik} , the subdeterminants from two rows of point coordinates. Analogously, in the n -ary domain we have rows of variables p_ρ , whose individual elements are the subdeterminants from ρ rows x , and rows of symbols S^ρ , whose elements behave like subdeterminants from ρ rows s cogredient to the u . The symbolic representation by determinants is known to hold only when the symbol rows S^ρ are resolved into the rows s and these are distributed among different determinants; only such aggregates of determinants as depend on the S^ρ then have real meaning. But which aggregates of determinants accomplish this cannot be surveyed, and thereby all the other theorems mentioned in the ternary domain are lost.

I would now like to state a simple principle by which one obtains a symbolic representation explicit in the rows S^ρ , and thereby also recovers the remaining theorems. This is an application of the familiar theorem on corresponding matrices: “To each row of variables p_ρ there corresponds one-to-one another row $q_{n-\rho}$ whose individual elements are subdeterminants from $n - \rho$ rows u connected with the ρ rows x by equations $(u | x) = 0$.” Correspondingly, to each symbol row S^ρ there is assigned another $S_{n-\rho}$, behaving as if it were composed from $n - \rho$ rows cogredient to the x .

The theorem also permits a second formulation, suited for applications: “To every matrix product $(v^{(1)} \dots v^{(\rho)} | y^{(1)} \dots y^{(\rho)})$,⁴ where the rows of the second matrix are contragredient to those of the first, there is assigned a determinant; and conversely each determinant can be transformed into a matrix product with a prescribed number ρ of rows.” Thus determinant and matrix product appear as fully equivalent, and the first theorem of the symbolic method assumes the form: “All invariant formations can be represented symbolically by matrix products.” From two symbol rows S^σ , T^τ one can now, with the aid of rows of variables, very easily form a matrix product explicitly containing these symbol rows and hence representing an invariant formation of the form $(S^\sigma | p_\sigma)(T^\tau | p_\tau)$, namely the following:

$$(q_{n-\sigma-\lambda} S^\sigma | T_{n-\tau} p_{\tau-\lambda}), \quad (\lambda = 1, 2, \dots, \tau \text{ or } n - \sigma). \quad (1)$$

More important is the converse: that all invariant formations can be represented explicitly by matrix products, so that the above process is the extension of the binary and ternary folding process. To prove this converse it is necessary to transfer the second theorem of the symbolic method to the n -ary domain, namely to set up the totality of the independent indecomposable identities in which all rows S^ρ , p_ρ are regarded as indecomposable. These identities arise from the known ones containing only rows x and u ,⁵ by replacing the

¹Lecture delivered at the Salzburg meeting of natural scientists, 1909.

²Study: *Methoden zur Theorie der ternären Formen*. II. § 6. (Leipzig, Teubner, 1889.)

³Über die Bildung des Formensystems der ternären biquadratischen Form. § 1. (Crelle’s Journal Bd. 134. 1908.)

⁴That is, the sum over the products of corresponding determinants, formed from the matrix of the ρ rows v and that of the ρ rows y .

⁵E. Pascal in *Memorie d. R. Acc. d. Lincei*. (4) 5 (1888), p. 375.

product theorem for determinants by a system of product theorems for matrices, and by contracting different matrix products into a single one by means of precisely these product theorems. One obtains in this way two dualistic formulas, of which only one is given:

$$(S_1^{\rho_1} S_2^{\rho_2} \cdots S_k^{\rho_k} | xp_{\rho-1}) = \sum_{i=1}^k \varepsilon (S_1^{\rho_1} \cdots S_{i-1}^{\rho_{i-1}} S_{i+1}^{\rho_{i+1}} \cdots S_k^{\rho_k} q_{n-\rho_i+1} | S_{n-\rho_i} x),$$

$$\rho = \sum_{i=1}^k \rho_i < n, \quad \varepsilon = \pm 1. \quad (2)$$

The only specialization contained in these formulas, namely that a row x enters, cannot be avoided; for completely general symbol and variable rows we arrive at a “decomposition identity,” an identity in which a row p_ρ is decomposed into the individual rows x . The simplest special case of the decomposition identity was already used by Clebsch in his “fundamental problem of invariant theory” to derive the elementary covariants in the theory of series expansions. With the aid of these identities, which mediate the transition from the non-explicit determinant expressions to the matrix product, one can in fact show that the desired explicit symbolic representation is obtained with the matrix representation.

For the folding process, however, a theorem entirely analogous to that in the ternary domain holds, and the reduction theorems can likewise be built upon it analogously. If in (1) one denotes the number λ as the defect of the folding, and calls foldings for which $\sigma = \tau$ cogredient, then all foldings can be generated by successive application of the cogredient foldings of defect 1. The proof of this theorem rests essentially on the fact that the most general forms can be replaced by “normal forms,” that is, by forms that vanish identically under the application of all invariant differential operations. In the quaternary domain this latter fact was proved by Mertens;⁶ our knowledge of the most general differential operations—which, as is well known, arise from the folding processes by replacing the symbol rows by differential symbols—allows us, using the decomposition identity, to transfer Mertens’s proof almost verbatim to the n -ary domain.

⁶Über invariante Gebilde quaternärer Formen. Wiener Berichte. Bd. 98. 1889.

4. On the Invariant Theory of Forms in n Variables

Journal für die reine und angewandte Mathematik 139 (1911), pp. 118–154

Introduction.

In the projective invariant theory of forms in n variables, the questions adjoining the main problem—the proof of finiteness of the system of forms—have scarcely been treated when one goes beyond the binary and ternary domains. These are the questions concerning the mutual dependence of the forms and the methods for controlling this connection among forms. These methods rest essentially—as has been completely proved in the binary and ternary domains—on two theorems designated by Study as the “fundamental theorems of the symbolic method,” namely the theorem on the symbolic representability of the forms and the theorem on symbolic identities.⁷ The latter theorem says that all relations existing among integral invariant formations are obtained by successive application of a small number of symbolic identities; thus the symbolic method, and only it, gives knowledge of the connection among forms without leaving the domain of invariants.

Gordan, in his Erlangen program,⁸ already pointed to the reason for the difficulty in passing to n variables: it lies in the fact that the first theorem of the symbolic method is no longer valid in its general form. In the n -ary domain there occur, as is well known, variable rows of higher level p_ρ ⁹ and analogously symbol rows of higher level S^ρ ; one arrives at these rows both when, as originally with Clebsch,¹⁰ one starts from forms with arbitrarily many rows p_ρ , and when one starts, following F. Deruyts and A. Capelli,¹¹ from forms containing only arbitrarily many cogredient rows x . A symbolic representation that would contain the rows p_ρ , S^ρ explicitly¹² does not exist; it is necessary to resolve the p_ρ , S^ρ into the individual rows x and the s contragredient to them, and to distribute these among different determinants. It is indeed possible, by means of differential conditions (cf. formula (19.)), to verify that a given aggregate of determinants has the property of depending only on the rows p_ρ , S^ρ ; but in this way one obtains no overview of the totality of invariant formations and of the processes for their generation.¹³

We shall now show in §§ 2 and 6 how, using the “theorem on corresponding matrices,”¹⁴ the first fundamental theorem is recovered in its generality by replacing representation by determinants with representation by “matrix products.” In § 2 it is shown that every matrix product explicitly containing the rows S^ρ , p_ρ is an invariant formation of the ground form; the converse given in § 6, that every invariant formation can be represented explicitly by matrix products, succeeds only through the transfer of the second fundamental theorem (§§ 3–5).

This theorem is known for forms that contain only variable rows x .¹⁵ The general problem, to set up the totality of the indecomposable identities in which all rows S^ρ , p_ρ are regarded as indecomposable, was formulated by Study;¹⁶ at the same time he sees in the number of identities that occur a measure of the difficulty of the methods in the n -ary domain. Since it is now possible to combine the totality of the identities in a single formula (36.), this difficulty is at least pressed down to a minimum. In applications, however, certain distinguished special cases of (36.) will often occur, such as (20.), (21.), characterized by the fact that they admit explicit representation without substitution of new rows; or formula (31.), designated as the “decomposition identity,” since a row p_ρ appears to be decomposed into the rows y .

⁷E. Study, *Methoden zur Theorie der ternären Formen* (Leipzig, Teubner, 1889). II. § 6. For the occurrence of these fundamental theorems also in the invariant theory of special transformation groups, cf. Study, *Das Apollonische Problem* (Math. Ann. Bd. 49 [1897]) and *Zur Differentialgeometrie der analytischen Kurven* (Transactions of the American Math. Society, Bd. 1 [1909]).

⁸P. Gordan, *Über das Formensystem binärer Formen*. S. 46. (Leipzig, Teubner, 1875).

⁹For the notation see § 1.

¹⁰A. Clebsch, über eine Fundementalaufgabe der Invariantentheorie. (Abh. der Ges. d. Wiss. zu Göttingen, Bd. XVII, 1872).

¹¹Cf. A. Capelli, *Lezioni sulla teoria delle forme algebriche* (Napoli 1902), § 22–24, and the literature cited there.

¹²More precise definition of “explicit representation” in § 2.

¹³A computationally very valuable development of the symbolism due to Study (communicated by F. Engel, Ber. d. K. Sächs. Ges. d. Wiss., math.-phys. Kl., 1900, p. 126, and for the quaternary domain by E. Study, *Geometrie der Dynamen*, p. 135) is also not an explicit representation in our sense. It would, for example, hardly lead to the nonsymbolic differentiation processes for generating the forms.

¹⁴Apart from Grassmann’s *Ausdehnungslehre* of 1862, no. 112, first in A. Brill, über zwei Berührungsprobleme, § 2 (Math. Ann. Bd. 4. 1871).

¹⁵E. Pascal, Giorn. di mat. 26 (1888), pp. 33, 102; Rendic. d. Accad. d. Linc. (4) 4 (1888), p. 119; ib. Mem. (4) 5 (1888), p. 375.

¹⁶“Methoden . . .”, p. 204, note 17).

Upon the two fundamental theorems the further theory can now easily be built. The first theorem gives directly the processes for generating the forms, the symbolic folding process and the nonsymbolic differentiation processes,¹⁷ while the decomposition identity eliminates, among these processes, all equivalent ones (§ 6). Knowledge of the differentiation processes further yields the transfer of the concept of “normal form” to the n -ary domain and, by means of a proof procedure given by Mertens¹⁸ in the quaternary domain, the theorem that a system of invariant formations can be replaced by an equivalent system of normal forms (§ 7). Under this latter assumption I showed in my dissertation¹⁹ that the ternary folding process can be generated by successive application of two fundamental foldings. On the basis of this theorem and the theory of series expansions, I then developed, by introducing the concept of the “form row,” the principal reduction theorems for systems of forms. Entirely analogously, in the n -ary domain the folding process can be generated from $n - 1$ fundamental foldings (§ 8), and the reduction theorems can be set up (§ 9).

Afterword. On the occasion of the Salzburg communication, Prof. E. Müller of Vienna drew my attention to the fact that the calculation with matrix products underlying this paper is, in principle, identical with the calculus of Grassmann’s theory of extension.²⁰ Indeed, Grassmann’s “combinatorial product” $[S^{\rho_1} S^{\rho_2} \dots S^{\rho_i}]$ can be regarded as a matrix given by these rows (cf. § 2), specifically the “progressive product” as the matrix of the rows themselves, the “regressive” as the matrix of the assigned rows $S_{n-\rho_1}, S_{n-\rho_2}, \dots, S_{n-\rho_i}$, while the matrix corresponding to the “mixed product” arises when several rows S are combined by substitution (9.) into new rows P, Q . Our “assigned row” corresponds to Grassmann’s “complement,” our “matrix product” to a “mixed product of level number 0.”

Under this correspondence, the development given in the *Ausdehnungslehre* of 1862, no. 113, becomes identical with a formula dualistic to our formula (32.). In the special case $\rho = 1$, (32.) goes over into (17.), the dualistic formula into (16.). From these special cases of Grassmann’s development, Müller²¹ has recently derived the identities (20.), (21.) mentioned above.

In contrast to the Grassmann–Müller derivation, our derivation at the same time gives the proof of completeness of the identities, which appears to be the essential point for the questions considered here, and it proceeds from (20.), (21.) further to the most general indecomposable identities (36.), which do not seem to have been noticed before now.

§ 1. Notations and definitions. Summary of known results.

We denote, as usual, by the variable row x the row of elements $x_1, x_2, \dots, x_n$, and analogously by the variable row p_ρ ($\rho = 1, 2, \dots, n - 1$) the row of the $\binom{n}{\rho}$ elements $p_{i_1 i_2 \dots i_\rho}$, where $p_{i_1 i_2 \dots i_\rho}$ denotes the $\binom{n}{\rho}$ determinants formed from the matrix of the ρ rows $x^{(1)}, x^{(2)}, \dots, x^{(\rho)}$,

$$\begin{vmatrix} x_{i_1}^{(1)} & x_{i_2}^{(1)} & \cdots & x_{i_\rho}^{(1)} \\ x_{i_1}^{(2)} & x_{i_2}^{(2)} & \cdots & x_{i_\rho}^{(2)} \\ \vdots & \vdots & & \vdots \\ x_{i_1}^{(\rho)} & x_{i_2}^{(\rho)} & \cdots & x_{i_\rho}^{(\rho)} \end{vmatrix}, \quad (i_1 < i_2 < \cdots < i_\rho = 1, 2, \dots, n).$$

According to the theorem on corresponding matrices,²² to every row p_ρ there is assigned one-to-one a second row $q_{n-\rho}$ by the relation, valid for each element of the row,

$$p_{i_1 i_2 \dots i_\rho} = (-1)^{\frac{\rho(\rho+1)}{2} + i_1 + i_2 + \dots + i_\rho} q_{j_1 j_2 \dots j_{n-\rho}}, \quad (1)$$

¹⁷In the paper “Invariante Gebilde von Nullsystemen” (Sitzungsber. d. Ak. d. Wiss. Wien, Bd. 97, Abt. IIa, 1888), Mertens arrived by another path, not directly generalizable, at the quaternary folding process. The alternating invariant of 6 rows S^2 occurring there differs from the one arising with us by a single application of substitution (11.) by an aggregate of even invariants. For a special case the quaternary folding process is also found in P. Gordan, *Das simultane System von zwei quadratischen quaternären Formen* (Math. Ann. Bd. 56. 1903).

¹⁸F. Mertens, Über invariante Gebilde quaternärer Formen. (Sitzber. d. Ak. d. Wiss. Wien. Bd. 98. Abt. IIa. 1889.)

¹⁹Über die Bildung des Formensystems der ternären biquadratischen Form. (Dieses Journal, Bd. 134. 1908.)

²⁰Hermann Graßmanns gesammelte mathematische und physikalische Werke. Ersten Bandes zweiter Teil: Die Ausdehnungslehre von 1862. In Gemeinschaft mit H. Graßmann d. Jüngeren herausgegeben v. Fr. Engel (Leipzig, Teubner 1896).

²¹E. Müller, Beiträge zur Graßmannschen Ausdehnungslehre. 1. Mitteilung: Einige allgemeine Sätze. (Sitzungsber. d. Ak. d. Wiss. Wien; Bd. 118. Abt. IIa. Juli 1909), Satz 16 und 18.

²²Cf. for example Gordan–Kerscheneiner, *Vorlesungen über Invariantentheorie*, Bd. I, p. 99.

where the i and j are each in good order and together form a complete permutation of the numbers 1 through n . The row $q_{n-\rho}$ consists of the $\binom{n}{\rho}$ determinants of degree $n - \rho$, $q_{j_1 j_2 \dots j_{n-\rho}}$, formed from the matrix of $n - \rho$ rows $u^{(1)}, u^{(2)}, \dots, u^{(n-\rho)}$ contragredient to the x . These rows satisfy the $\rho(n - \rho)$ condition equations

$$(u^{(k)} | x^{(l)}) = \sum_{\alpha=1}^n u_{\alpha}^{(k)} x_{\alpha}^{(l)} = 0; \quad (k = 1, 2, \dots, n - \rho; l = 1, 2, \dots, \rho). \quad (2)$$

We shall assume them normalized (with Clebsch²³) so that the proportionality factor otherwise entering in (1.) is set equal to one.

According to Clebsch²⁴ (and likewise according to the later investigations of F. Deruyts and A. Capelli, cf. the Introduction), the most general forms can be reduced to those that contain at most one of each of the $n - 1$ rows p_{ρ} , and hence can be represented symbolically as follows:

$$(S^1 | p_1)^{m_1} (S^2 | p_2)^{m_2} \dots (S^{n-1} | p_{n-1})^{m_{n-1}}, \quad (3)$$

where, in analogy with (2.), we have set

$$(S^{\rho} | p_{\rho}) = \sum S^{i_1 i_2 \dots i_{\rho}} p_{i_1 i_2 \dots i_{\rho}}.$$

In consequence of the nonlinear identical relations existing between the elements of a row p_{ρ} , the symbol rows S^{ρ} can be normalized so that they satisfy the very same identities, hence so that they behave like the determinants of a ρ -rowed matrix whose rows $s^{(1)}, s^{(2)}, \dots, s^{(\rho)}$ are contragredient to the x .²⁵ Thus to each row S^{ρ} there can be assigned one-to-one another $S_{n-\rho}$ through the relation

$$S_{i_1 i_2 \dots i_{n-\rho}} = (-1)^{\frac{(n-\rho)(n-\rho+1)}{2} + i_1 + i_2 + \dots + i_{n-\rho}} S^{j_1 j_2 \dots j_{\rho}}, \quad (4)$$

where again the i and j are each in good order and together exactly exhaust the numbers 1 through n . The elements $S_{i_1 i_2 \dots i_{n-\rho}}$ of the row $S_{n-\rho}$ behave like the determinants formed from $(n - \rho)$ rows $\sigma^{(1)}, \sigma^{(2)}, \dots, \sigma^{(n-\rho)}$ cogredient to the x , and the rows s and σ are linked by the $\rho(n - \rho)$ conditions

$$(s^{(k)} | \sigma^{(l)}) = 0, \quad (k = 1, 2, \dots, \rho; l = 1, 2, \dots, n - \rho). \quad (5)$$

In consequence of the relations (1.) and (4.), each factor of (3.) is capable of the equivalent fourfold symbolic representation

$$(S^{\rho} | p_{\rho}) = (S^{\rho} q_{n-\rho}) = (S_{n-\rho} p_{\rho}) = (-1)^{\rho(n-\rho)} (q_{n-\rho} | S_{n-\rho}), \quad (6)$$

where, as always in what follows, $(S^{\rho} q_{n-\rho})$ and $(S_{n-\rho} p_{\rho})$ are abbreviations for the determinants $(s^{(1)} \dots s^{(\rho)} u^{(1)} \dots u^{(n-\rho)})$ and $(\sigma^{(1)} \dots \sigma^{(n-\rho)} x^{(1)} \dots x^{(\rho)})$; by Laplace expansion these prove to depend only on the rows $S^{\rho}, q_{n-\rho}$ and respectively $S_{n-\rho}, p_{\rho}$, as the above notation is meant to indicate. The expressions $(S^{\rho} | p_{\rho})$ and $(q_{n-\rho} | S_{n-\rho})$ mean, according to the preceding discussion, the products of the matrices of the rows s and x , respectively of the rows u and σ , where by matrix product is meant the sum over the products of corresponding determinants, that is, the value of the determinant of the product matrix.

The characteristic number $\rho(n - \rho)$ belonging to a symbol row (respectively variable row) will be called, following Clebsch, the weight of the row;²⁶ the number ρ , which gives the number of rows s , the upper weight index; the number $n - \rho$, which gives the number of rows σ , the lower weight index. It follows from the relations (4.) that a symbol row can occur in one and the same relation once with upper and once with lower weight index; the same holds for the occurrence of the corresponding rows p_{ρ} and $q_{n-\rho}$. It remains to note that we generally denote variable rows of higher level whose associated matrix consists of rows x by p , those whose matrix consists of rows u by q ; symbol rows cogredient to the x by small Greek letters $\sigma, \tau, \alpha, \beta$; symbol rows cogredient to the u by small Latin letters s, t, a, b ; all other symbol rows by capital Latin letters with weight index: $S^{\rho}, T^{\tau}, A^{\rho}$ or $S_{n-\sigma}, T_{n-\tau}, A_{n-\rho}$. Corresponding to an upper or lower weight index, we also write the individual elements of a row with upper or lower indices, as for instance in (4.).

²³a. a. O. § 5.

²⁴a. a. O. § 12.

²⁵Clebsch, a. a. O. § 4.

²⁶a. a. O. § 11.

§ 2. Representation by matrix products.

In what follows we always understand by “matrix product” the sum over the products of corresponding determinants of two matrices with the same number of rows, where the rows of the second matrix are contragredient to those of the first. The rows of each matrix may be: 1. individually given; 2. combined, as in (6.), into a row S^ρ ; 3. united into different rows $S^\rho, S^\sigma \dots$. The matrix product is to be denoted by the symbol $(|)$ used in (6.). The representation (6.) is a special application of a second formulation of the theorem on corresponding matrices: “To every ρ -rowed matrix product $(v^{(1)}v^{(2)} \dots v^{(\rho)} | y^{(1)} \dots y^{(\rho)})$ there is assigned a determinant; and conversely to every determinant there can be assigned a matrix product with prescribed number ρ of rows ($\rho = 1, 2, \dots, n-1$).”²⁷ Indeed, one has only to combine the rows y into a row p_ρ (respectively the rows v into a row q_ρ) and to carry out the substitutions given by (1.) in order to pass from a given matrix product to the determinant and conversely. The value of the determinant is, however, not given by its individual elements, but first results from Laplace expansion. Since determinant and matrix product therefore prove to be equivalent, the theorem on the symbolic representability of the forms (the first fundamental theorem) is expressed also as follows:

Theorem I. “All invariant formations can be represented symbolically by matrix products, and conversely all matrix products are invariant formations.”²⁸

Representation by matrix products allows us to overcome the essential difficulty of the non-explicit representation caused by the determinant representation.²⁹ By “explicit representation” we mean a representation into which only the rows $S^\rho, T^\tau, \dots$ themselves enter, or rows R^ρ uniquely derivable from these, but in which the individual component rows $s, t, \dots$ no longer occur. We shall obtain in § 6, for all invariant formations that depend only on the symbol rows $S^\rho, T^\tau, \dots$, a representation explicit in these symbol rows, and thereby the transfer to the n -ary domain of the generating processes, the folding process and the differentiation processes. The possibility of explicit representation is evident from the fact that only the simplest matrix products can be assigned precisely to the determinants occurring in the determinant representation, so that all other matrix products are combinations of different determinants into a single expression. For if one resolves the symbol and variable rows into the individual rows s, u, x , then, according to the first fundamental theorem in its usual form, an aggregate of determinants must necessarily arise. The proof that, conversely, all determinant aggregates that represent invariant formations of prescribed ground forms can be combined into explicit matrix products follows by using the second fundamental theorem (§ 6).

In order to obtain from two symbol rows S^σ, T^τ , with the help of variable rows, an invariant formation of the ground form $(S^\sigma | p_\sigma)(T^\tau | p_\tau)$ that explicitly contains all rows (the folding process), we need only form a matrix product of the third kind according to the definition given at the beginning of the paragraph:

$$(q_{n-\sigma-\lambda}S^\sigma | T_{n-\tau}p_{\tau-\lambda}), \quad (\lambda = 1, 2, \dots, \tau, n-\sigma). \quad (7)$$

Expanded, (7.) gives:

$$\begin{aligned} & (v^{(1)} \dots v^{(n-\sigma-\lambda)} s^{(1)} \dots s^{(\sigma)} | \alpha^{(1)} \dots \alpha^{(n-\tau)} y^{(1)} \dots y^{(\tau-\lambda)}) \\ &= \sum_{i,k,l} \varepsilon q_{k_1 \dots k_{n-\sigma-\lambda}} S^{k_{n-\sigma-\lambda+1} \dots k_{n-\lambda}} T_{l_1 \dots l_{n-\tau}} p^{l_{n-\tau+1} \dots l_{n-\lambda}}, \end{aligned} \quad (8)$$

that is, an expression depending only on the rows S, T, q, p . Here $\varepsilon = \pm 1$ ³⁰ and the sum is to be understood in such a way that the indices of each row $S \dots$ are in good order, and that for each combination $i_1, i_2, \dots, i_{n-\lambda}$ the k as well as the l individually run through all then possible permutations of the numbers i . The number λ entering into (7.) will be called the defect of the folding in the case $\sigma > \tau$; the reason for this latter restriction appears in § 6.³¹

The determinants of the two matrices entering in (7.) can be regarded as elements of new rows $Q^{n-\lambda}, P_{n-\lambda}$ whose associated rows are Q_λ, P^λ . According to (8.), these rows Q, P can be expressed through the original S and q , respectively T and p . We indicate this by the equations

$$\begin{aligned} q_{n-\sigma-\lambda}S^\sigma &= Q^{n-\lambda} \sim Q_\lambda, & \text{or also } Q_\lambda &= [q_{n-\sigma-\lambda}S^\sigma]_\lambda, \\ T_{n-\tau}p_{\tau-\lambda} &= P_{n-\lambda} \sim P^\lambda, & \text{or also } P^\lambda &= [T_{n-\tau}p_{\tau-\lambda}]^\lambda. \end{aligned} \quad (9)$$

²⁷For $\rho > n$ the matrix product is known to vanish; for $\rho = n$ it goes into the product of two determinants.

²⁸Invariant formations of prescribed ground forms are, of course, only those aggregates of matrix products that depend on the symbol rows $S^\rho \dots$

²⁹Cf. the Introduction.

³⁰This notation is to be retained throughout.

³¹The number λ always runs up to the smaller of the two last indicated numbers.

Taking (4.) into account, one then has, analogously to (6.), the equivalent fourfold symbolic representation for the matrix product (7.):

$$(q_{n-\sigma-\lambda}S^\sigma | T_{n-\tau}p_{\tau-\lambda}) = (q_{n-\sigma-\lambda}S^\sigma P^\lambda) = (Q_\lambda T_{n-\tau}p_{\tau-\lambda}) = (-1)^{\lambda(n-\lambda)}(P^\lambda | Q_\lambda). \quad (10)$$

A further substitution of new rows can be performed on (7.) by setting

$$(q_{n-\sigma-\lambda}S^\sigma | T_{n-\tau}p_{\tau-\lambda}) = (R^{\sigma+\lambda} | p_{\sigma+\lambda})(R^{\tau-\lambda}p_{\tau-\lambda}). \quad (11)$$

Here too (8.) shows that the products of the elements of the rows R can be expressed uniquely through the rows S and T . The substitutions (9.) and (11.) will be used especially when a further folding with other rows is performed with (7.).

The matrix representation enables us to state invariantly the quadratic relations between the elements of a row p_ρ (from which, as is known, all the others follow), hence, according to § 1, the relations linear in the coefficients of the ground form that the rows S^ρ satisfy. These are the identities

$$(q_{n-\rho-\lambda}S^\rho | S_{n-\rho}p_{\rho-\lambda}) = 0, \quad (\lambda = 1, 2, \dots, \rho, n - \rho), \quad (12)$$

where the p and q are to be regarded as arbitrary quantities. We shall see in § 5 that the identities with odd defect number λ are consequences of those with higher defect. The relations (12.) are a direct consequence of the relations (5.); for one has

$$(q_{n-\rho-\lambda}S^\rho | S_{n-\rho}p_{\rho-\lambda}) = (v^{(1)} \dots v^{(n-\rho-\lambda)} s^{(1)} \dots s^{(\rho)} | \sigma^{(1)} \dots \sigma^{(n-\rho)} y^{(1)} \dots y^{(\rho-\lambda)}),$$

and the application of the product theorem gives, in consequence of (5.), a vanishing $(n-\lambda)$ -rowed determinant. The identities (12.) say that in order to find all types of foldings it suffices to fold the product of forms linear in each variable row; or also, by § 6, that the normalized form $(S^\rho | p_\rho)^m$ possesses no invariant formation linear in the coefficients.³²

The conditions (12.) are also sufficient to characterize the rows p_ρ . For the property that the individual elements of p_ρ are determinants of a matrix is an invariant property, and therefore it must be expressible invariantly; but by Theorem VI, the conditions (12.) give the greatest possible invariant restriction for the p_ρ , namely that the linear form formed from the row p_ρ as coefficients possesses no invariant formations at all.

§ 3. The symbolic identities.

We now have to transfer the second fundamental theorem of the symbolic method, namely to set up the totality of the irreducible indecomposable identities in which all rows S^ρ, p_ρ are regarded as indecomposable. We also stipulate the following. Corresponding to the meaning of symbol and variable rows, those identities are to be regarded as composite that arise when one specializes the variable row p_ρ to $p_{\rho-1}y$ and applies one of the identities of the system to the resulting expressions; while, on the other hand, identities containing k symbol rows are to count as indecomposable even if they arise from identities with fewer symbol rows by the above process. The rows S^ρ, p_ρ need not necessarily enter the identities explicitly; for their interpretation as indecomposable quantities it suffices to show that the matrix products that occur depend only on these rows. We shall show, however, that in this case, partly by applying substitution (11.), an explicit representation can always be obtained. The general identities are obtained, by means of the principle of matrix representation, from the special identities given by Pascal, which contain only rows x and u and constitute a direct transfer of those given by Study for the ternary domain:³³

$$\sum_{i=1}^n (-1)^{i+1} (a^{(i)} | x) (a^{(1)} \dots a^{(i-1)} a^{(i+1)} \dots a^{(n)} u) = (-1)^{n+1} (u | x) (a^{(1)} \dots a^{(n)}), \quad (13)$$

$$\sum \pm (a^{(1)} | x^{(1)}) (a^{(2)} | x^{(2)}) \dots (a^{(n)} | x^{(n)}) = (a^{(1)} \dots a^{(n)}) (x^{(1)} \dots x^{(n)}), \quad (14)$$

$$\sum_{i=1}^n (-1)^{i+1} (u | a^{(i)}) (a^{(1)} \dots a^{(i-1)} a^{(i+1)} \dots a^{(n)} x) = (-1)^{n+1} (u | x) (a^{(1)} \dots a^{(n)}). \quad (15)$$

³²Cf. an addendum by Study in Grassmann's works, first volume, second part, p. 510 (condition that a quantity S^ρ is simple).

³³For formulation and literature of the problem, cf. the Introduction.

Two further identities arise from (13.) and (15.) through the substitutions $(a^{(i)} | x) = (a^{(i)} q_{n-1})$ and respectively $(u | a^{(i)}) = (p_{n-1} a^{(i)})$, and so by our interpretation they are not to be regarded as essentially different. (14.) represents the product theorem for determinants; we shall show how (13.) and (14.) (and, dualistically, (15.) and (14.)) can be replaced by the system of suitably formed product theorems for matrices. The product theorem, formed once for ρ -rowed matrices and once for $(\rho - 1)$ -rowed matrices, reads

$$(a^{(1)} a^{(2)} \dots a^{(\rho)} | x p_{\rho-1}) = (a^{(1)} a^{(2)} \dots a^{(\rho)} | x y^{(2)} \dots y^{(\rho)}) = \sum \pm (a^{(1)} | x) (a^{(2)} | y^{(2)}) \dots (a^{(\rho)} | y^{(\rho)}),$$

$$\sum \pm (a^{(2)} | y^{(2)}) \dots (a^{(\rho)} | y^{(\rho)}) = (a^{(2)} \dots a^{(\rho)} | y^{(2)} \dots y^{(\rho)}) = (a^{(2)} \dots a^{(\rho)} p_{\rho-1}) = (a^{(2)} \dots a^{(\rho)} q_{n-\rho+1});$$

and from this follows the system of product theorems, where $\rho = 2, 3, \dots, n$:

$$\begin{aligned} (a^{(1)} a^{(2)} \dots a^{(\rho)} | x p_{\rho-1}) &= \sum_{i=1}^{\rho} (-1)^{i+1} (a^{(i)} | x) (a^{(1)} \dots a^{(i-1)} a^{(i+1)} \dots a^{(\rho)} | p_{\rho-1}) \\ &= \sum_{i=1}^{\rho} (-1)^{i+1} (a^{(i)} | x) (a^{(1)} \dots a^{(i-1)} a^{(i+1)} \dots a^{(\rho)} q_{n-\rho+1}). \end{aligned} \quad (16)$$

For $\rho = n$, (16.) goes over into (13.); if we further specialize $p_{n-1} = y^{(2)} p_{n-2}$, $p_{n-2} = y^{(3)} p_{n-3}$, etc., and apply the identities (16.) to the expressions $(a^{(1)} \dots a^{(i-1)} a^{(i+1)} \dots a^{(n)} | y^{(2)} p_{n-2})$, $(a^{(1)} \dots a^{(i-1)} a^{(i+1)} \dots a^{(n-1)} a^{(n+1)} \dots a^{(n)} | y^{(3)} p_{n-3})$, etc., then we pass from (13.) to (14.). Thus the identity (14.) is, by our definition, composite from the identities (16.); equivalently, the system (16.) is equivalent to the identities (13.) and (14.). The system (16.) satisfies one of the conditions required for the general identities: it contains the row $p_{\rho-1}$ as an indecomposable quantity and at the same time in explicit form. It is the only system satisfying this condition and only this condition. For we can form no further determinants or matrix products from the rows a, x, p ; and between these no relations independent of (16.) can exist, since otherwise, by eliminating $(a^{(1)} \dots a^{(\rho)} | x p_{\rho-1})$, there would arise a relation between ρ ($\rho < n$) expressions $(a^{(i)} | x) (a^{(1)} \dots a^{(i-1)} a^{(i+1)} a^{(\rho)} q_{n-\rho+1})$, whereas by (13.) at least $n + 1$ such expressions must enter into a relation. From analogous considerations one obtains the system equivalent to (15.) and (14.):

$$\begin{aligned} (u q_{\rho-1} | a^{(1)} \dots a^{(\rho)}) &= \sum_{i=1}^{\rho} (-1)^{i+1} (u | a^{(i)}) (q_{\rho-1} | a^{(1)} \dots a^{(i-1)} a^{(i+1)} \dots a^{(\rho)}) \\ &= \sum_{i=1}^{\rho} (-1)^{i+1} (u | a^{(i)}) (p_{n-\rho+1} a^{(1)} \dots a^{(i-1)} a^{(i+1)} \dots a^{(\rho)}). \end{aligned} \quad (17)$$

In order to pass from (16.) to identities between arbitrary rows $A^{\rho_1}, B^{\rho_2}, \dots, x, p$, we observe that these identities must become identical with (16.) as soon as we resolve $A^{\rho_1} = a^{(1)} a^{(2)} \dots a^{(\rho_1)}$, $B^{\rho_2} = b^{(1)} b^{(2)} \dots b^{(\rho_2)}$, etc., and that they can also be combined into final expressions only by the operations determined by (16.) as soon as the individual expressions are of the type entering into (16.). Thus we obtain, if we also set

$$\rho = \rho_1 + \rho_2 + \dots + \rho_k \leq n, \quad (18)$$

$$\begin{aligned} (a^{(1)} \dots a^{(\rho_1)} b^{(1)} \dots b^{(\rho_2)} \dots k^{(1)} \dots k^{(\rho_k)} | x p_{\rho-1}) &= (A^{\rho_1} B^{\rho_2} \dots K^{\rho_k} | x p_{\rho-1}) \\ &= \sum_{i=1}^{\rho_1} (-1)^{i+1} (a^{(i)} | x) (a^{(1)} \dots a^{(i-1)} a^{(i+1)} \dots a^{(\rho_1)} B^{\rho_2} \dots K^{\rho_k} q_{n-\rho+1}) \\ &\quad + (-1)^{\rho_1 \rho_2} \sum_{i=1}^{\rho_2} (-1)^{i+1} (b^{(i)} | x) (b^{(1)} \dots b^{(i-1)} b^{(i+1)} \dots b^{(\rho_2)} A^{\rho_1} \dots K^{\rho_k} q_{n-\rho+1}) \\ &\quad + \dots \\ &\quad + (-1)^{(\rho_1 + \dots + \rho_{k-1}) \rho_k} \sum_{i=1}^{\rho_k} (-1)^{i+1} (k^{(i)} | x) (k^{(1)} \dots k^{(i-1)} k^{(i+1)} \dots k^{(\rho_k)} A^{\rho_1} \dots K^{\rho_{k-1}} q_{n-\rho+1}). \end{aligned}$$

Each individual sum (but not already a part of the sum) does in fact contain the rows $A^{\rho_1}, B^{\rho_2}, \dots$ as indecomposable quantities; for it satisfies the necessary and sufficient conditions³⁴

$$\left(a^{(i+1)} \left| \frac{\partial}{\partial a^{(i)}} \right. \right) = 0; \dots; \left(k^{(j+1)} \left| \frac{\partial}{\partial k^{(j)}} \right. \right) = 0. \quad (i = 1, 2, \dots, \rho_1 - 1; j = 1, 2, \dots, \rho_k - 1) \quad (19)$$

We show that it also contains all rows explicitly. For if we set $B^{\rho_2} \dots K^{\rho_k} q_{n-\rho+1} = q_{n-\rho_1+1}$, then by (16.) and (10.) we obtain

$$\begin{aligned} \sum (-1)^{i+1} (a^{(i)} | x) (a^{(1)} \dots a^{(i-1)} a^{(i+1)} \dots a^{(\rho_1)} q_{n-\rho+1}) &= (A^{\rho_1} | x p_{\rho_1-1}) \\ &= (A_{n-\rho_1} x p_{\rho_1-1}) = (-1)^{(\rho_1+1)(n+1)} (q_{n-\rho_1+1} | A_{n-\rho_1} x) \\ &= (-1)^{(\rho_1+1)(n+1)} (B^{\rho_2} \dots K^{\rho_k} q_{n-\rho+1} | A_{n-\rho_1} x). \end{aligned}$$

If we compute the remaining sums analogously and subsequently set, since the rows are already characterized as distinct by the weight indices, $B^{\rho_2} = A^{\rho_2}$, $C^{\rho_3} = A^{\rho_3}, \dots$, then we obtain the desired identities:

$$\begin{aligned} (A^{\rho_1} A^{\rho_2} \dots A^{\rho_k} | x p_{\rho-1}) &= \sum_{i=1}^k (-1)^{\delta_i} (A^{\rho_1} \dots A^{\rho_{i-1}} A^{\rho_{i+1}} \dots A^{\rho_k} q_{n-\rho+1} | A_{n-\rho_i} x), \\ \delta_i &= (\rho_1 + \rho_2 + \dots + \rho_{i-1}) \rho_i + (\rho_i + 1)(n + 1), \end{aligned} \quad (20)$$

and from (17.) the dualistic identities:

$$\begin{aligned} (u q_{\sigma-1} | A_{\sigma_1} A_{\sigma_2} \dots A_{\sigma_k}) &= \sum_{i=1}^k (-1)^{\varepsilon_i} (u A^{n-\sigma_i} | p_{n-\sigma+1} A_{\sigma_1} \dots A_{\sigma_{i-1}} A_{\sigma_{i+1}} \dots A_{\sigma_k}), \\ \varepsilon_i &= (\sigma_1 + \sigma_2 + \dots + \sigma_{i-1}) \sigma_i + (\sigma_i + 1)(n + \sigma). \end{aligned} \quad (21)$$

Theorem II. With (20.) and (21.) the totality of the indecomposable identities between the rows A^{ρ_i}, x, p , respectively A_{σ_i}, u, q , is exhausted, and at the same time the totality of those indecomposable identities that admit explicit representation without carrying out substitution (11.).

The first part of the theorem follows from the considerations leading to (20.) and (21.); the second part from the impossibility of an identity between two symbol rows:

$$(q_{\rho} A^{\sigma} | B_{\rho+\sigma-\tau} p_{\tau}) = \varepsilon_1 (q_{\rho} B^{n-\rho-\sigma+\tau} | A_{n-\sigma} p_{\tau}) + \varepsilon_2 (q_{\rho} q_{n-\tau} | B_{\rho+\sigma-\tau} A_{n-\sigma}),$$

where $\rho \leq \tau \leq \sigma$ and none of the rows has weight $1 \cdot (n - 1)$. (Other assumptions about ρ, τ, σ can be reduced to this one by interchanging the row designations.) This impossibility follows, for example, by expanding (8.) under the specialization $q_{\rho} = l M^{\rho-1}$, $q_{n-\tau} = l N^{n-\tau-1}$, if one sets $l_1 = 1$, $M^{2,3,\dots,\rho} = 1$, $A^{\rho+1,\dots,\rho+\sigma} = 1$, all other elements of the rows l, M, A equal to zero, while N and B are arbitrary. Since identities between arbitrary rows, by resolving a row p_{τ} into $y^{(1)} y^{(2)} \dots y^{(\tau)}$, pass into identities composite from (20.), the second part of the assertion is proved as soon as (20.) can be generated from identities with only two symbol rows. In fact, from

$$(A^{\rho_1} A^{\rho_2} | x p_{\rho-1}) = \varepsilon (A^{\rho_1} q_{n-\rho+1} | A_{n-\rho_2} x) + (A^{\rho_1} | x p_{\rho_1-1}), \quad \text{where } p_{\rho_1-1} \sim A^{\rho_2} q_{n-\rho+1}, \quad (20a)$$

by specializing $A^{\rho_1} = B^{\sigma_1} B^{\sigma_2}$, that is, by

$$(B^{\sigma_1} B^{\sigma_2} | x p_{\rho_1-1}) = \varepsilon (B^{\sigma_1} q_{n-\rho_1+1} | B_{n-\sigma_2} x) + (B^{\sigma_1} | x p_{\sigma_1-1}), \quad \text{where } p_{\sigma_1-1} \sim B^{\sigma_2} q_{n-\rho_1+1},$$

one obtains an identity (20.) in three symbol rows; and by further specializations of $B^{\sigma_1} = C^{\tau_1} C^{\tau_2}$, etc., the general identities (20.). The nonexistence of the explicit representation of an identity between two symbol rows and two arbitrary variable rows thus implies the nonexistence in the case of arbitrarily many symbol rows, provided that among the variable rows no row x or u is contained.

³⁴Capelli, a. a. O. § 23, gives sufficient conditions: $(a^{(k)} | \frac{\partial}{\partial a^{(i)}}) = 0$, ($k > i$, $i = 1, 2, \dots, \rho_1 - 1$). These most simply imply the necessary and sufficient conditions by application of the Poisson bracket process after Wellstein, Von den Differentialgleichungen der projektiven Invarianten, Math. Ann. 67 (1909) § 3.